5G MAG The Media Connectivity Association



OVERVIEW AND UPDATES · CONNECTED MEDIA PRODUCTION

Network APIs for Connectivity Quality

Requesting, monitoring, and managing dynamic network resources and QoS

CAMARA Project

NOVEMBER 2026 – www.5g-mag.com

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About the project

The documents the project rests on

CAMARA APIs

Open source project

Portable network APIs: a media application requests network behaviour, for example a QoS boost for a live stream, through common interfaces.

3GPP TS 29.522

Technical Specification

The NEF northbound APIs that the CAMARA QoS requests map onto, with Npcf_PolicyAuthorization in TS 29.514.

3GPP TS 23.434

Technical Specification

SEAL, for network capability exposure to applications; introduced in Release 16. Release 20 (work plan of 28 September 2026): SEAL Phase 4, content unverified.

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The APIs analysed

The CAMARA APIs for connectivity quality that 5G-MAG analyses for media

Request quality QoS Profiles are consulted first by most of these

QoS Profiles | Quality on Demand | QoS Provisioning | QoS Booking | QoS Booking and Assignment

Reserve a network for an area and a time window

Dedicated Networks | Network Slice Booking

Check and monitor against an Application Profile

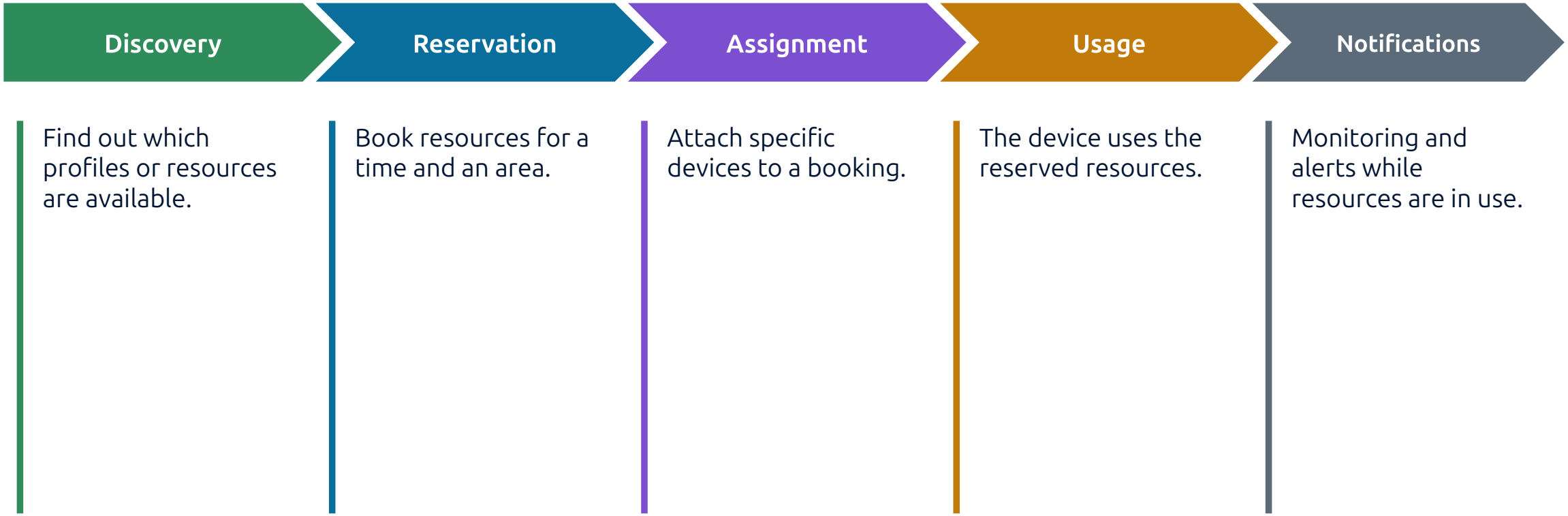
Application Profiles | Connectivity Insights | Connectivity Insights Subscriptions

Quality on Demand applies a profile to one session; QoS Provisioning keeps it on a device until deleted; QoS Booking reserves it for a future window.

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Five phases

The phases the analysis uses to compare what each API does



Phases as in the Workflows analysis. Each API covers some of them; Dedicated Networks covers discovery, reservation and assignment.

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What you can build

The two work areas of the analysis, over mobile networks

Single device

Production and contribution

One camera or contribution feed set up at short notice, such as mobile journalism, newsgathering or uplink video: immediacy and simplicity.

Many devices

Production and contribution

A coordinated production, such as outside broadcast or remote production, with many devices and flows that may each need a different QoS.

Live distribution

Live media distribution

Live TV and radio to many devices. Content provider and operator share what each can observe, to diagnose degraded delivery.

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Repository

On main, as an API client: Dedicated Networks portal and MCP server, and Insomnia collections for the QoS APIs

REPOSITORY	WHAT IT IS	COVERS	LATEST RELEASE	
rt-camara-examples	example files and tools for the CAMARA Project: a Dedicated Networks portal, an MCP server and Insomnia collections	Management portal, MCP server, Insomnia collections	v1.0.0 28 Oct 2026	GitHub

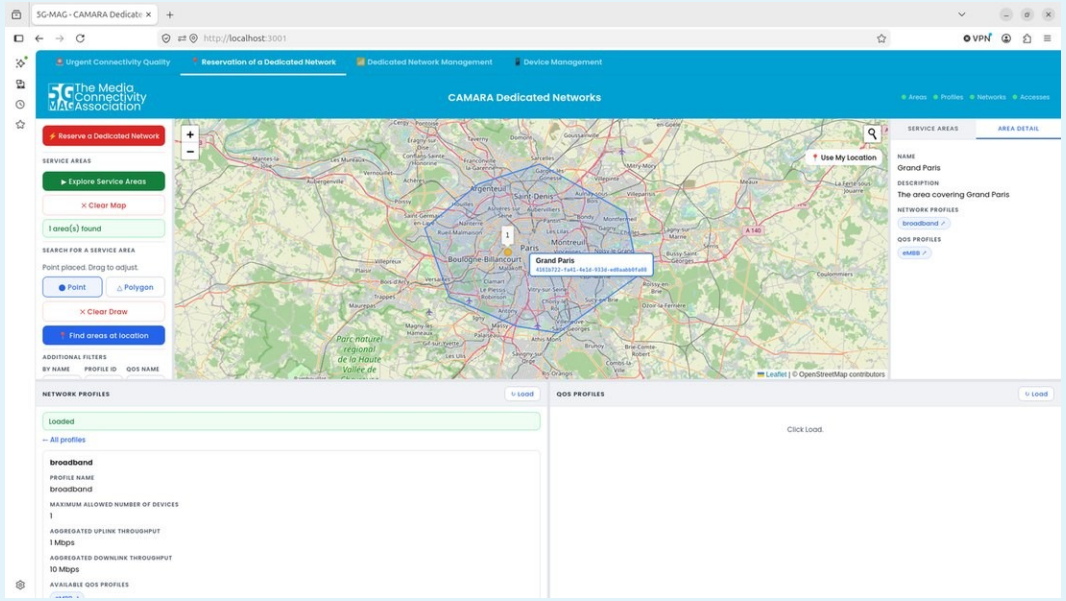
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● released ● no release yet

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See it and try it

The Dedicated Networks management portal, and the application showcase demo built on it



Dedicated Networks management portal



Application showcase demo

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WHERE WE STAND

Network APIs for Connectivity Quality

API

BROUGHT TO STUDY	ON-GOING	OUTCOMES
Alignment with CAMARA Sync26 meta-release: Dedicated Networks r2.3 (rc planned); code follows r2.2 (rc)	Initial implementation of Quality on Demand (API) consumer with Telefónica QoD sandbox	Insomnia collections for Dedicated Networks Dedicated Networks portal for reservation and device access MCP Server for AI assistants

RELATION WITH SDOS AND INDUSTRY FORA

CAMARA Project

CAMARA Industry survey on CQM APIs for Media Production

CAMARA Contribution to the paper "CAMARA CQM APIs: Understanding Connectivity Quality Management"

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